Chipsee

Industrial PC

# EPC/PPC-A8-080-R




PN: CS80600T080

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# Contents


**1 Product Overview** 3

**2 Ordering Options** 4

2.1 Operating System 4

2.2 Optional Features 4

**3 Hardware Features** 5

**4 Power Input** 6

**5 Touch screen** 7

**6 Connectivity** 7

6.1 RS232/RS485/CAN 7

6.2 USB Connectors 9

6.3 LAN Connectors 9

6.4 WiFi & BT Module 10

6.5 3G/4G/LTE Module 10

6.6 GPIO Port 11

**7 TF Card Slot** 13

**8 Audio Connectors** 13

**9 Boot DIP Switch** 14

**10 Mounting Procedure** 14

**11 Mechanical Specifications** 14

11.1 EPC-A8-080-R 14

11.2 PPC-A8-080-R 15

**12 3D Model** 16

**13 Disclaimer** 16

**14 Technical Support** 16

# EPC/PPC-A8-080-R



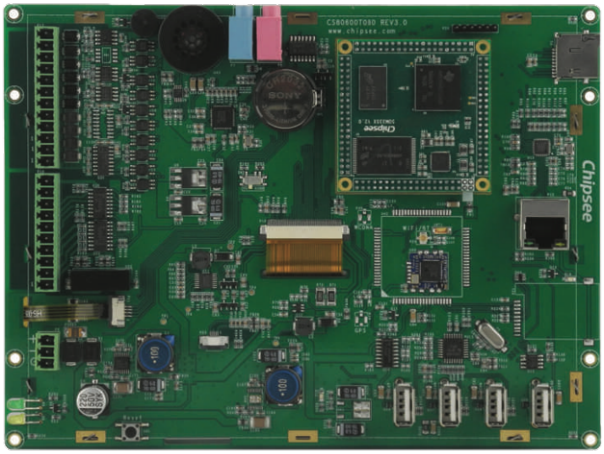

Front View




Rear View




Front View (Embedded Variant)




Rear View (Embedded Variant)

## 1 Product Overview

The Cortex®-A8 series EPC/PPC-A8-080-R (PN: CS80600T080) is a high-quality industrial PC. This single board computer features a 8.0” resistive touch screen with a resolution of 800 x 600 pixels and a brightness of 250 cd/$m^2$.

Key Applications

- Human Machine Interface  HMI
- Process Control
- Process Monitoring
- HMI
- Infotainment
- Predictive Maintenance
- Machine Learning
- Machine Vision
- Automotive applications

- Gaming...

It is available both as an embedded solution and as a device hosed in an aluminum casing with bezels, thus facilitating different installation options:

- Installation on an industrial cabinet
- Integration with the existing equipment

The EPC/PPC-A8-080-R Industrial PC is based around the powerful CS-SOM335X-V3 System on Module (SoM), powered by the AM3354 Arm® Cortex®-A8 microprocessor unit (MPU). The AM33554 MPU is enhanced with image, graphics processing, peripherals and industrial interface options such as EtherCAT and PROFIBUS. The devices support high-level operating systems (HLOS).

The microprocessor unit (MPU) subsystem is based on the ARM Cortex-A8 processor and the PowerVR SGX™ Graphics Accelerator subsystem provides 3D graphics acceleration to support display and gaming effects.

The AM33554 MPU does not generate extensive heat, so even the thin aluminum housing on PPC version delivers sufficient thermal dissipation.

# 2 Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Hardware Features section provides information about the default options bundled with the product.

> **Note**
>
> You can order EPC/PPC-A8-080-R from the official Chipsee Store or from your nearest distributor.

## 2.1 Operating System

This product comes with a pre-installed OS of your choice. Please see the list below for the supported OSes, which can be also obtained from the Software Documentation section, along with the detailed installation instructions.

- Android 4.1
- Linux with Qt 4.8
- Linux with Qt 5.5
- Debian 7.4
- Debian 8.4
- Angstorm v2012.12

> **Warning**
>
> The Software Documentation section provides a detailed instruction how to install different OS on your own. However, bear in mind that Chipsee can't take the responsibility of inadequate installation procedure. If you "brick" your device, please contact Chipsee Technical Support at **support@chipsee.com** for further assistance

## 2.2 Optional Features

The EPC/PPC-A8-080-R Industrial PC does not include WiFi/BT and/or 3G/4G modules by default. Both the Wi-Fi/BT and 3G/4G LTE module module are optional and can be selected during the ordering process.

> **Warning**
>
> Installation, repair, and maintenance tasks should be performed by trained personnel only.
> Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

# 3 Hardware Features

The EPC/PPC-A8-080-R Industrial PC offers a board range of performance and connectivity options for scalable integration, providing expandability according to future needs. Some of the key features are listed in the table below.

**Key Features**

| EPC/PPC-A8-080-R | |
|---|---|
| **CPU** | AM3354ZCZ100, Arm® Cortex®-A8, 1GHz |
| **RAM** | 512MB DDR3 |
| **eMMC** | 4GB |
| **Storage** | TF Card, Supports up to 32GB SDHC |
| **Display Size** | 8.0” |
| **Display Type** | LCD |
| **Display Resolution** | 800 x 600 |
| **Brightness** | 250 $cd/m^2$ |
| **Touch Technology** | Resistive |
| **USB** | 4 x USB 2.0 Host |
| **LAN** | 1 x Channel 100M LAN |
| **Audio** | 3.5mm output/input connector, 2W Internal Speaker |
| **Buzzer** | Yes |
| **RTC** | Yes |
| **RS232** | 2 x RS232 |
| **RS485** | 2 x RS485[1] |
| **CAN** | 1 x CAN[1] |
| **GPIO** | 8 Channels, 4 Input, 4 Output, 1000V DC Isolated |
| **WiFi/BT** | Onboard WiFi/BT **(optional)** |
| **Expansion Port** | N/A |
| **3G/4G/LTE** | Onboard 3G **(optional)** |
| **Power Input** | From 6V to 42V |
| **Current at 12V** | 400mA Max |
| **Power Consumption** | 6W Typical |
| **Working Temperature** | From -20°C to +70°C |
| **OS** | Android 4.1, Linux with Qt 4.8, Linux with Qt 5.5, Debian 7.4, Debian 8.4, Angstorm v2012.12 |

| EPC/PPC-A8-080-R | |
|---|---|
| **Dimensions** | EPC-A8-080-R (PN: CS80600T080E): 199 x 149 x 29mm |
| | PPC-A8-080-R (PN: CS80600T080P): 226 x 175 x 31mm |
| **Weight** | EPC-A8-080-R (PN: CS80600T080E): 580g |
| | PPC-A8-080-R (PN: CS80600T080P): 1070g |
| **Mounting** | EPC-A8-080-R (PN: CS80600T080E): Embedded |
| | PPC-A8-080-R (PN: CS80600T080P): Panel |

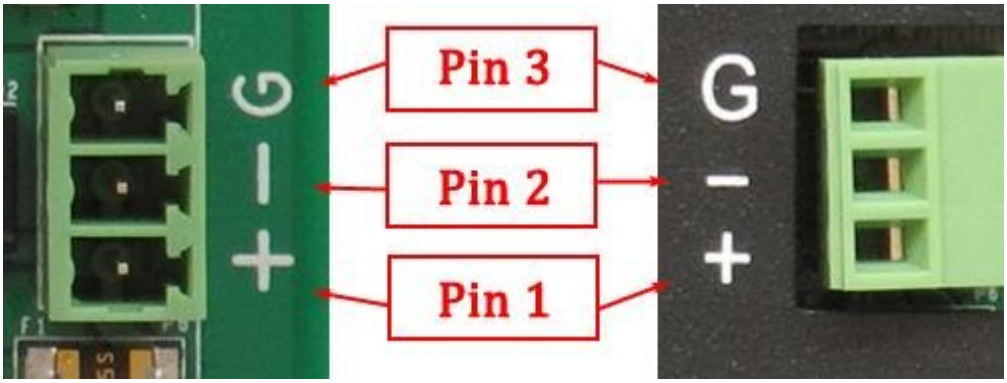
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[1] (1,2)
The RS485 and CAN channels may be customized to the following arrangement:

- 2 x RS485, 1 x CAN (Default)
- 1 x RS485, 2 x CAN

# 4 Power Input

The EPC/PPC-A8-080-R Industrial PC can be powered by a wide range of input voltages: From 6V to 42V DC. The power input connector is a **3-pin, 3.81mm terminal**. The polarity and the pinout is clearly marked on the housing of the PPC version, as well as on the PCB itself of the EPC version, as shown in the figure below.




*Power Input (embedded/enclosed version)*

Note that the "**+**" sign represents the positive power input, and it is printed both at the casing and as a silk-screen on a PCB of the embedded version. The "**-**" terminal is shorted to the ground.

**Power Connector**

| Power Input Definition | | |
|---|---|---|
| **Pin Number** | **Definition** | **Description** |
| Pin 1 | Positive Input | DC Power **Positive Terminal** |
| Pin 2 | Negative Input | DC Power **Negative Terminal** |
| Pin 3 | Ground | **Power System Ground** |

> **Note**
>
> The system ground "**G**" is connected to power negative "**-**" on board.

# 5 Touch screen

EPC/PPC-A8-080-R is equipped with a resistive touch screen. The resistive touch screen is an ideal option for harsh industrial conditions due to its high immunity against high temperature, dirt, and dust. In addition, it does not require any special material to be operated; the resistive touch screen can be used with regular rubber gloves, fingers, or a plastic stylus.



*Resistive Touch Connector*

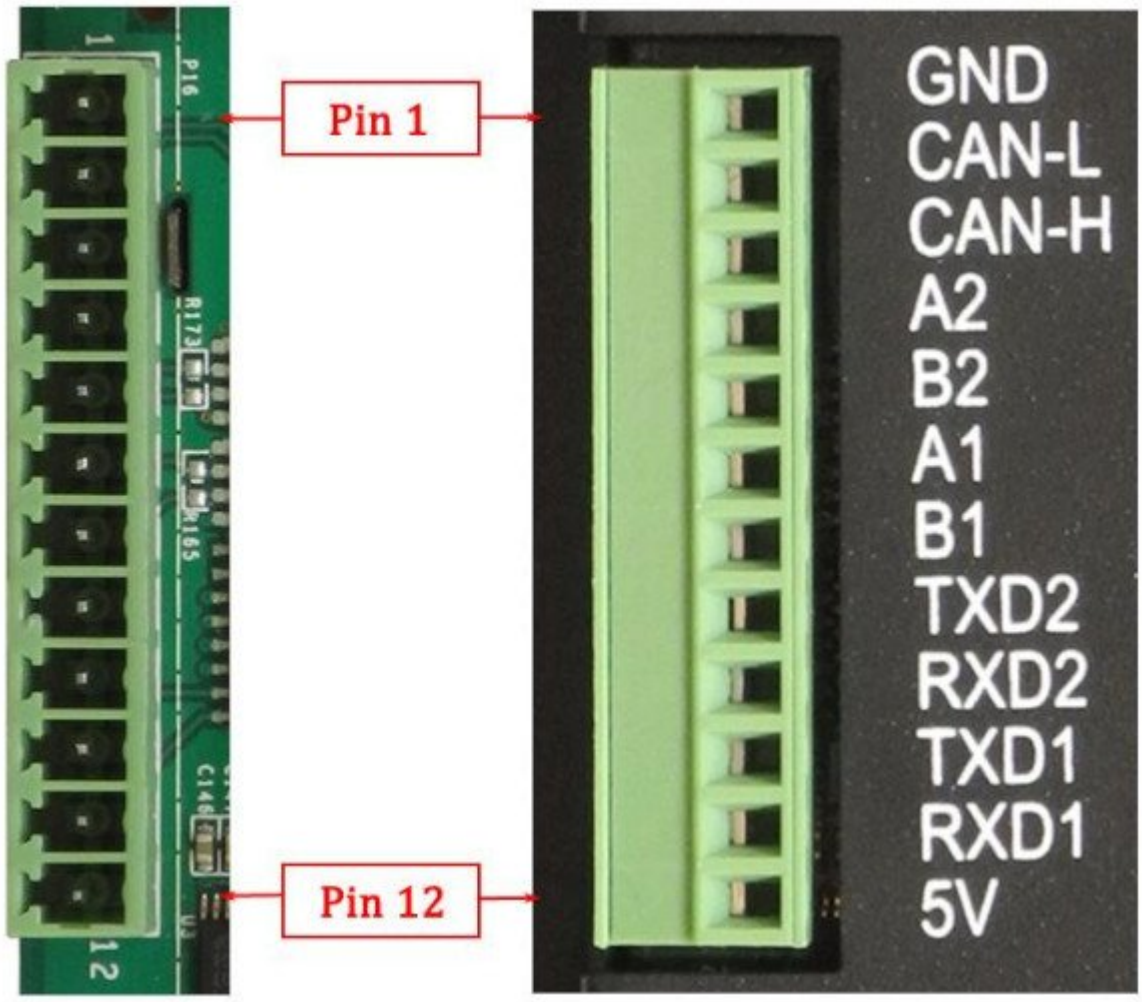
**Size/Type:** 8.0" high-temperature resistant, five-wire analog resistive touch screen
**Surface Hardness:** 3H
**Service Life (MTBF):** 35 million click events
**Light Transmittance:** > 81%

# 6 Connectivity

There are many connectivity options available on the EPC/PPC-A8-080-R industrial PC. It has 4 x USB 2.0 Host (can be customized to Host or OTG), 1 x Channel 100M LAN (RJ45) Ethernet connector supporting up to 1 Gbps, and 5 x UART terminals (RS232/RS485).

## 6.1 RS232/RS485/CAN

The serial communication interfaces (RS485, RS232, and CAN) are routed to a **12-pin 3.81mm terminal**, as illustrated on the figure below.




*Relation between serial pins on embedded vs. enclosed version of the* EPC/PPC-A8-080-R *Industrial PC*

The table below offers more detailed description of every pin and its definition:

**Connectivity Section**

| **RS232 / RS485 / CAN Pin Definition:** | | |
|---|---|---|
| **Pin Number** | **Definition** | **Description** |
| Pin 12 | +5V | System +5V Power Output, No more than 1A Current output |
| Pin 11 | RXD1 | UART0 of CPU, RS232 RXD Signal |
| Pin 10 | TXD1 | UART0 of CPU, RS232 TXD Signal |
| Pin 9 | RXD2 | UART1 of CPU, RS232 RXD Signal |
| Pin 8 | TXD2 | UART1 of CPU, RS232 TXD Signal |
| Pin 7 | B1 | UART2 of CPU, RS485 B Signal |
| Pin 6 | A1 | UART2 of CPU, RS485 A Signal |
| Pin 5 | B2 | UART4 of CPU, RS485 B Signal |
| Pin 4 | A2 | UART4 of CPU, RS485 A Signal |
| Pin 3 | CAN_H | DCAN0 of CPU, CAN H Signal |
| Pin 2 | CAN_L | DCAN0 of CPU, CAN L Signal |
| Pin 1 | GND | Isolated Ground Output |

## 6.2 USB Connectors

There are 4 x Type A **USB HOST connectors** onboard, as shown on the figure below.



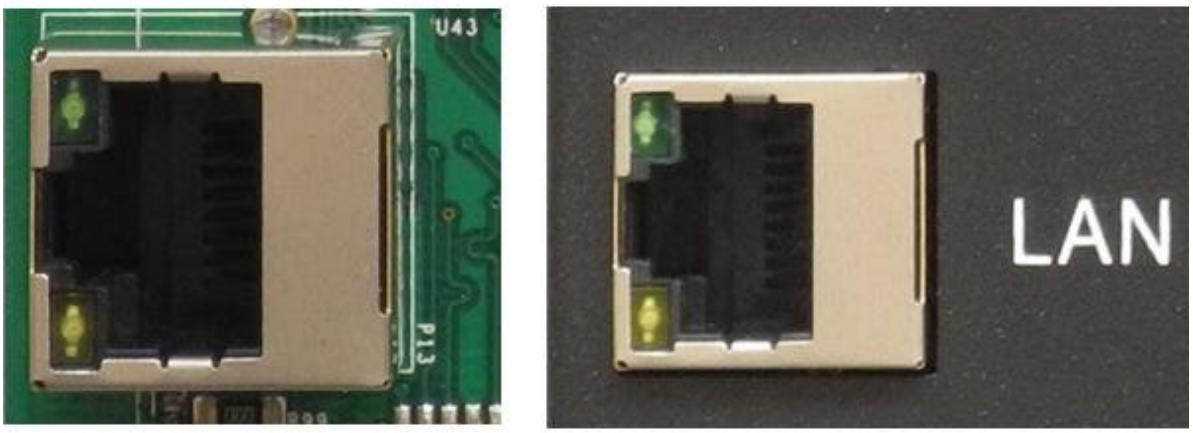
*USB HOST Connectors (embedded/enclosed PC version)*

> **Note**
>
> The USB Connectors are defined as HOST by default. If customer needs it work as OTG (slave), please solder a 0Ω 0603 Package Resistor to **R119** and **R121**.

> **Warning**
>
> Be careful not to touch surrounding electronic components accidentally while plugging in USB devices into the embedded Industrial PC version.

## 6.3 LAN Connectors

**LAN (RJ45) connector** provides Ethernet connectivity over standardized Ethernet cables as shown the figure below. The integrated Ethernet interface supports up to 1 Gbps data throughput.

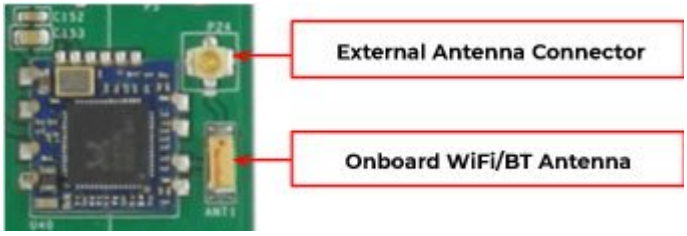



*RJ45 LAN Connectors (embedded/enclosed PC version)*

> **Note**
>
> Use CAT5 or better cables to achieve full data throughput over maximum distance defined by the 1000BASE-T standard (100m).

## 6.4 WiFi & BT Module

The EPC/PPC-A8-080-R Industrial PC is equipped with the popular **Realtek RTL8723 WiFi/BT module** that supports BT/BLE 4.0 (with backward compatibility), as well as 802.11bgn 2.4 GHz Wireless LAN (WLAN).




*RTL8273 WiFi/BT Module*

The enclosed (PPC) variant of the product also includes an SMA connector for an external WiFi/BT antenna, as illustrated in the figure below.




*WiFi+BT Antenna*

**Note**

1. The product does not come shipped with the WiFi/BT module by default.
2. If the operator mounts the WiFi/BT module on the EPC/PPC-A8-080-R industrial PC, the module uses the USB1 channel to communicate with CPU, so it will occupy the USB1 channel.

## 6.5 3G/4G/LTE Module

The EPC/PPC-A8-080-R Industrial PC is equipped with a **mini-PCIe connector** that can connect to a 3G/4G/LTE module. The customer will also need a SIM Card Holder and a 3G/4G/LTE Antenna Connector to ensure 3G/4G/LTE works on the EPC/PPC-A8-080-R.




*3G/4G/LTE Module*



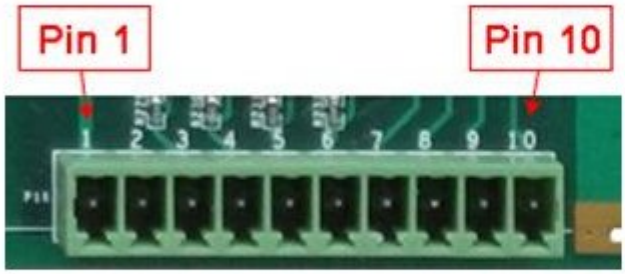


*SIM Card Holder and 3G/4G/LTE Antenna Connector*

> **Attention**
>
> 1. The product does not come shipped with the 3G/4G module by default.

## 6.6 GPIO Port

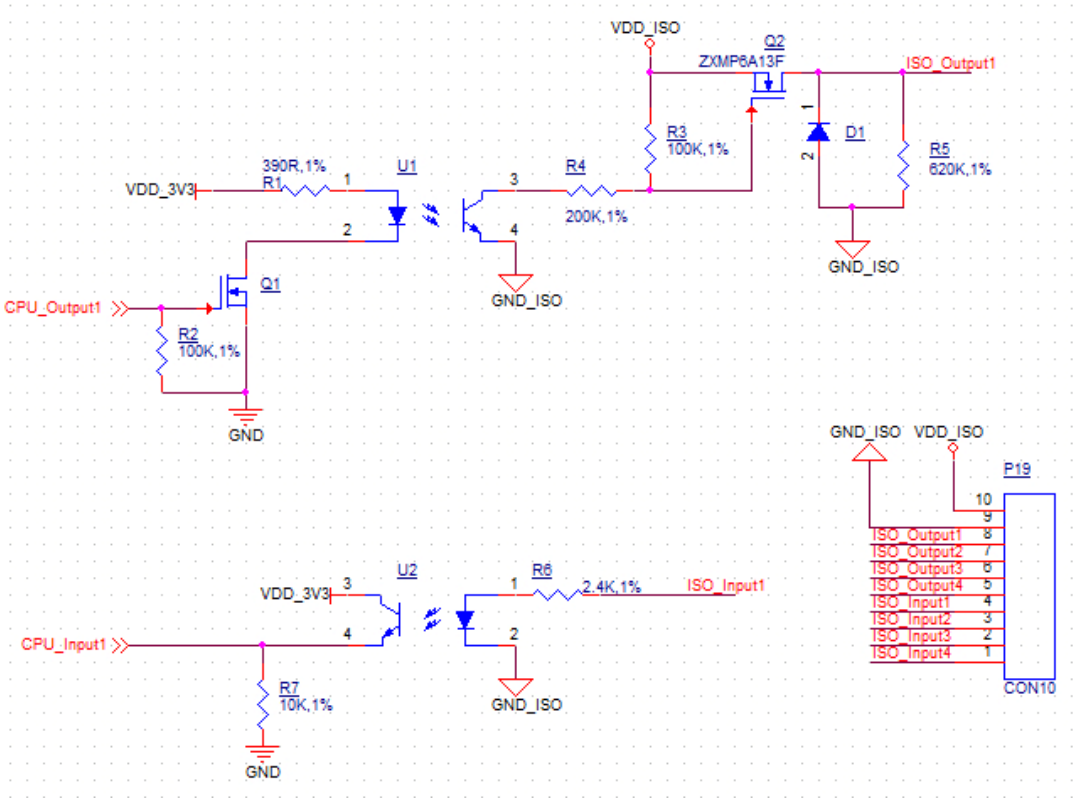
The EPC/PPC-A8-080-R Industrial PC has a 10 Pin 3.81mm **GPIO Connector**, as shown on the figure below, that is labeled as P18 on the PCB. The table below gives details about the definition of every Pin.







*GPIO Connector*




Isolated GPIO reduced schematic

### GPIO Connector Pin-out

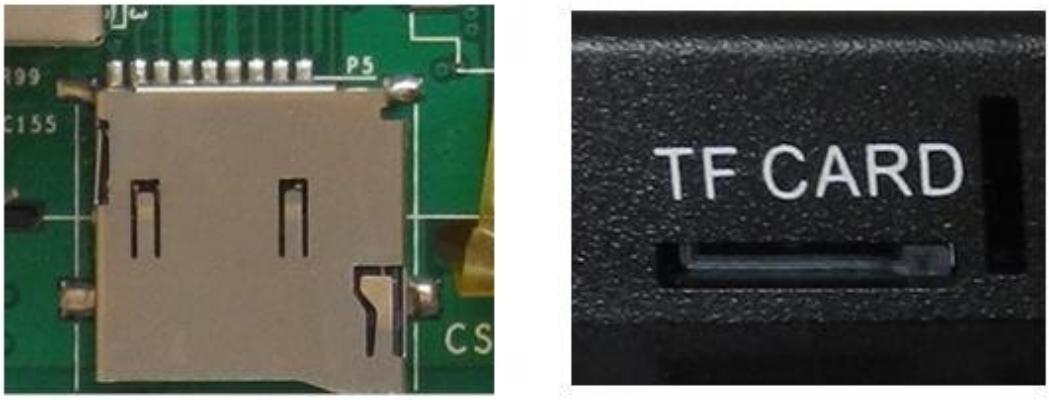
| GPIO Connector Pin Definition: | | |
|---|---|---|
| **Pin Number** | **Definition** | **Description** |
| Pin 1 | VCC | Isolated Power +5V Output |
| Pin 2 | GND | Isolated Ground |
| Pin 3 | OUT1 | Isolated Output 1 |
| Pin 4 | OUT2 | Isolated Output 2 |
| Pin 5 | OUT3 | Isolated Output 3 |
| Pin 6 | OUT4 | Isolated Output 4 |
| Pin 7 | IN1 | Isolated Input 1 |
| Pin 8 | IN2 | Isolated Input 2 |
| Pin 9 | IN3 | Isolated Input 3 |
| Pin 10 | IN4 | Isolated Input 4 |

**Attention**

- The GPIO has been Opt-Isolated and it uses the 24V Logic by default. The GPIO is driven by the on board VDD, you do not need an external isolated power input.
- The 4 output channels can drive at most 500mA current on each channel.
- The operator can use the power input that connects to Power Input Connector, if the operator doesn't want to use an isolated power input.
- Also, you can use the 5V voltage on-board as power input. The operator must solder a 0Ω resistor on R292 & R293 then you can connect the 5V voltage onboard to the **Isolated Power Input** Pin.

# 7 TF Card Slot

The EPC/PPC-A8-080-R Industrial PC features 1 x **TF Card (micro SD) slot**. It can address up to 32GB of memory.



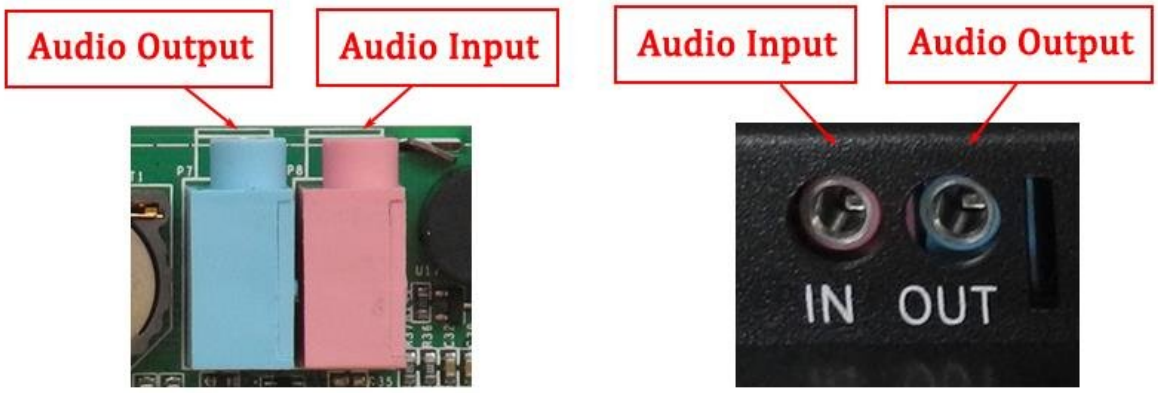

*TF (micro SD) Card Slot*

> **Note**
>
> The product does not come shipped with the TF Card by default.

# 8 Audio Connectors

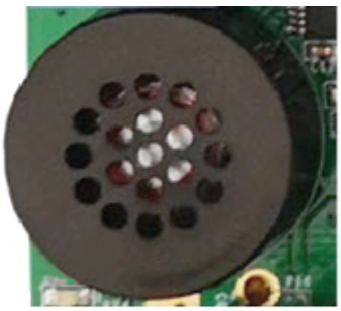
The EPC/PPC-A8-080-R Industrial PC features some audio peripherals, as well. It has 1 x **3.5mm audio input jack** and 1 x **3.5mm audio output jack**.

On the embedded panel PC version, the pink connector is the audio input jack (line-in) and the blue connector is the audio output jack (line-out, typically around -10 dBV). On the enclosed panel PC version, both audio input and audio output are clearly marked on the figure below.




*Audio I/O (embedded/enclosed PC version)*

In addition, EPC/PPC-A8-080-R features a miniature 2W embedded speaker for audio reproduction, as well as a small buzzer for alarm/notification sounds.



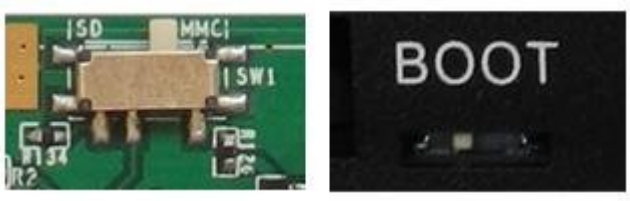


*2W Micro Speaker and Buzzer*

# 9 Boot DIP Switch

The EPC/PPC-A8-080-R Industrial PC supports boot from SD card. If you want to reflash the Operating System (OS), you can use the TF card for that purpose, combined with the **DIP switch** settings as illustrated in the figure below.

There is no need to alter the DIP switch settings during regular operation. However, if you need to reinstall the OS, please refer to the table below. Detailed information on how to re-flash the OS can be found in the Software Documentation.




*Boot DIP Switch*

**Boot Configuration Selection**

| Boot Config Select | | | | |
|---|---|---|---|---|
| **DIP SW** | **1** | **2** | **3** | **4** |
| **SD** | 1 | 0 | 0 | 0 |
| **eMMC** | 1 | 1 | 0 | 1 |
| **Download** | 0 | 1 | 1 | 0 |

# 10 Mounting Procedure

The EPC/PPC-A8-080-R Industrial PC can be mounted with 8 x M4 screws, enabling simplified installation onto any standard mounting fixture. Other mounting options might also be supported according to the table in the Hardware Features section.

You can find detailed information about mounting in the Mount IPC Guide.

# 11 Mechanical Specifications

## 11.1 EPC-A8-080-R

The outer mechanical dimensions of EPC-A8-080-R are 199 x 149 x 29mm (W x L x H).

## 11.2 PPC-A8-080-R

For PPC-A8-080-R, the outer mechanical dimensions are 226 x 175 x 31mm (W x L x H).

**Panel Mounting**




*Fixing* PPC-A8-080-R *industrial PC into panel*

**Note**

With the PPC-A8-080-R industrial PC, the operator can fix the PC into the panel by pushing it from the front inside the panel as described in the figure above. The recommended maximum thickness of the panel material is 8mm.

1. Make sure the Panel PC is configured correctly. The Boot Switch is sitting inside the housing. To use it, the Panel PC has to be unmounted from the panel.
2. Push the Panel PC straight into the Panel Hole until the unit sits flat on the panel as shown in the figure above.
3. Use the mounting fixtures to lock the Panel PC into it's place.

# 12 3D Model

EPC/PPC-A8-080-R 3D model can be viewed in the online doc in a web browser, **if you are reading from the PDF** version, please visit the online doc EPC/PPC-A8-080-R, select hardware documentation, drag the navigation bar to the 3D Model section.

# 13 Disclaimer

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# 14 Technical Support

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